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Unit -4

Trees: N-ary Trees, Binary Trees, Binary Search Trees, Expression Trees, Forests. Properties and conversion of N-ary tree / Forest to Binary tree

Presented By,
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Dept. of CSE,RVCE

TREES • N-ARY TREES

Why Do We Need Trees?

LINEAR

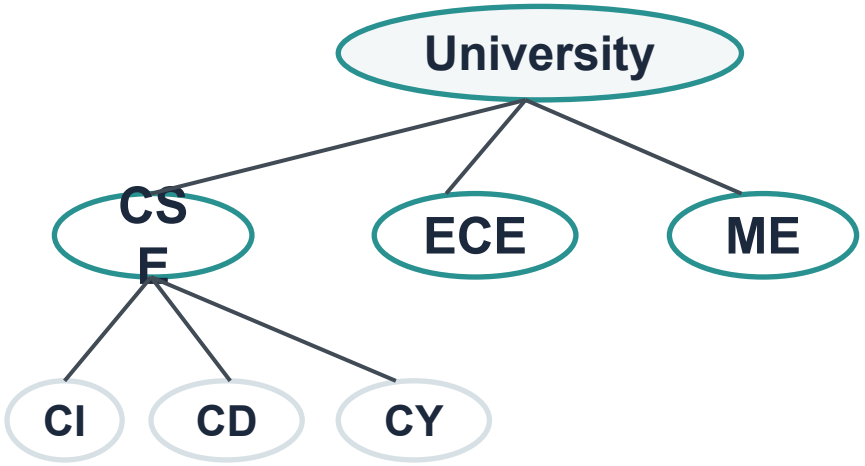
A → B → C → D

Data is mainly organized in a sequence.

Array • List • Stack • Queue

→

HIERARCHICAL



Real-world data often has PARENT → CHILD relationships.

Question: Which structure is better suited for a hierarchy?

TREES • N-ARY TREES

What Is a Tree?

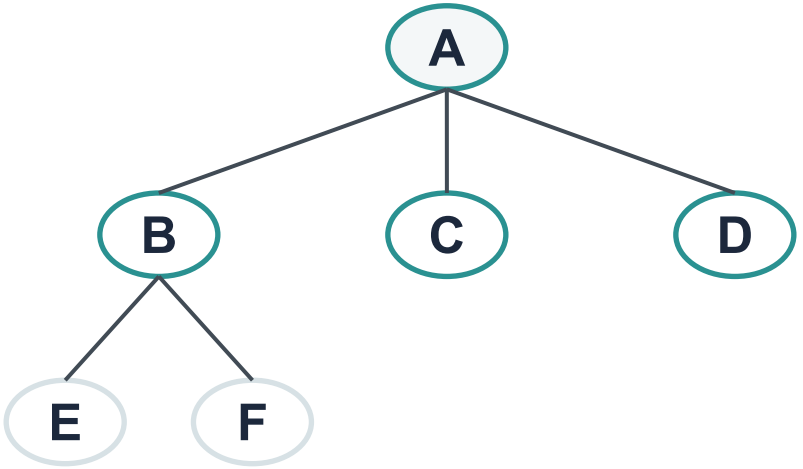
THE BASIC IDEA

A tree organizes nodes in a hierarchical form.

Nodes are connected by edges.
There is a topmost node called the root.

NON-LINEAR DATA STRUCTURE

A SIMPLE TREE



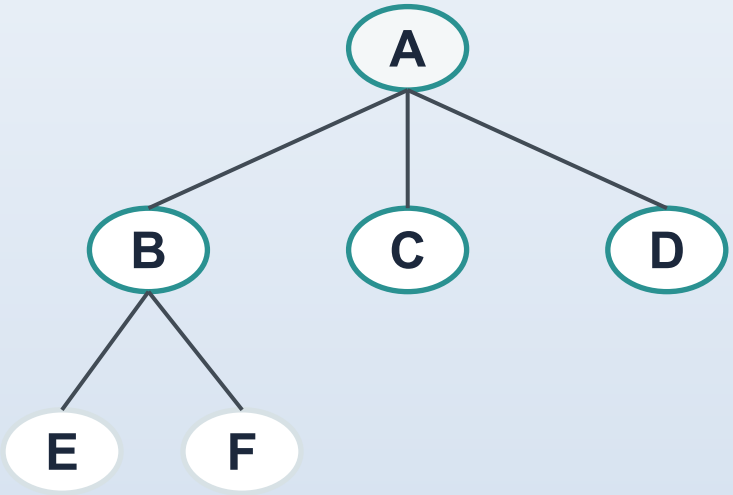
The structure grows downward from the ROOT.

TREES • N-ARY TREES

Understanding the Structure

Use one tree to learn the language of trees.

A = ROOT



B, C, D = CHILDREN

B, C, D = SIBLINGS

E, F = LEAVES

A = PARENT OF B

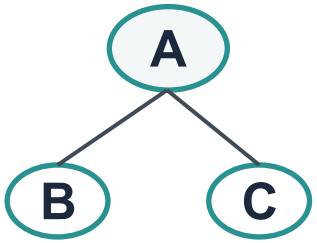
EDGE = CONNECTION BETWEEN TWO NODES

TREES • N-ARY TREES

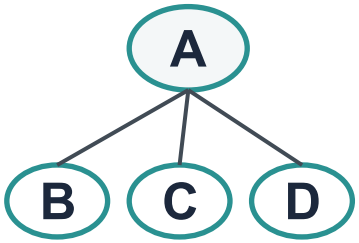
What Makes a Tree N-ary?

An N-ary tree allows each node to have AT MOST N CHILDREN.

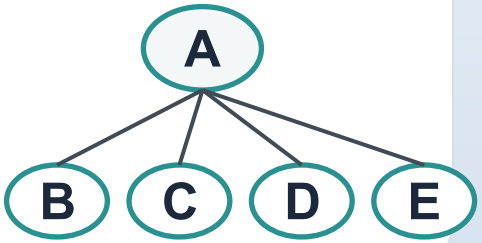
N = 2
BINARY



N = 3
TERNARY



N = 4
4-ARY



Important: “at most N” means a node can have 0, 1, 2, …, N children.

N-ary Tree — Definition

The value of N determines the maximum number of children a node can have.

An N-ary tree is a tree in which each node has AT MOST N CHILDREN.

0
Leaf node
No children

1
Single child
One child

2 … N
Internal node
Up to N children

Example: In a 3-ary tree, every node can have 0, 1, 2, or 3 children — never more than 3.

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Applications of N-ary Trees

Useful when a node can naturally have more than two children.

File & Folder Systems

A folder can contain many files and subfolders.

Menus & Navigation

A menu item can branch into several options.

DOM / HTML Structure

Elements form a hierarchy where a parent may have many child elements.

AI Search & Game Trees

A state can lead to multiple possible next states.

Organization Hierarchies

One manager or unit can have many direct subordinates.

Trie & Prefix Structures

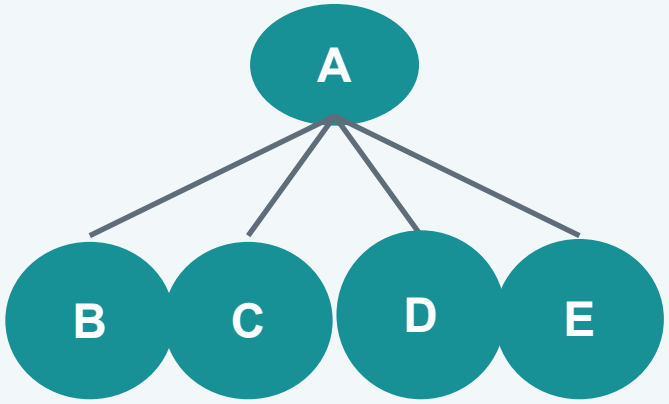
A node can branch to many possible symbols or characters.

KEY IDEA → N-ary trees model naturally branching relationships.

From N-ary Tree to Binary Tree

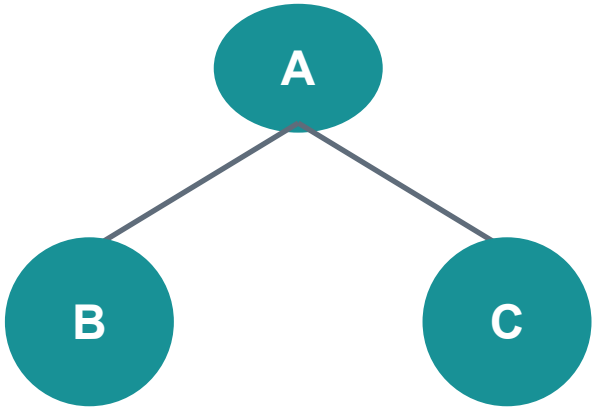
What happens when every node is allowed only two children?

N-ARY TREE



A can have many children

BINARY TREE



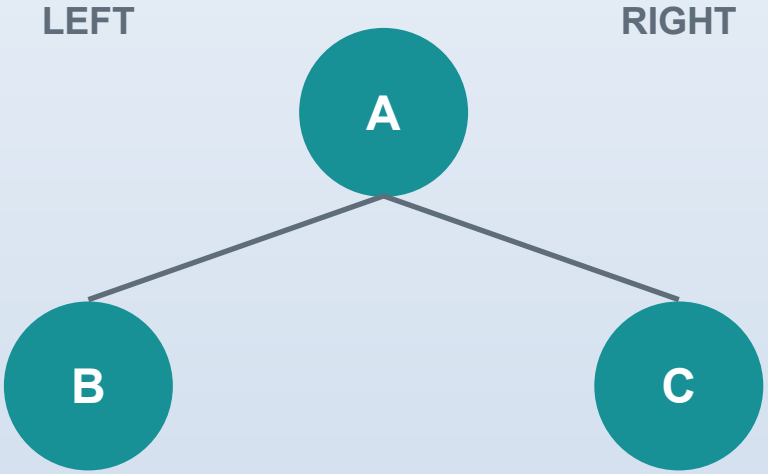
A has at most two children

KEY TRANSITION → Restrict the maximum children from N to 2.

What Is a Binary Tree?

A special type of tree with a strict limit on children.

A binary tree is a tree in which each node has AT MOST TWO CHILDREN, called the LEFT CHILD and RIGHT CHILD.

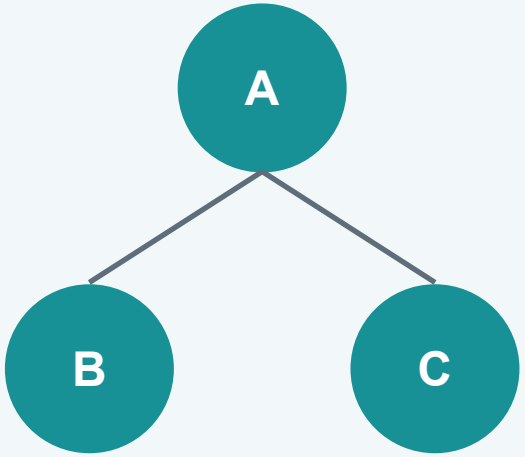


A node may have 0, 1, or 2 children — but never more than 2.

Left Child ≠ Right Child

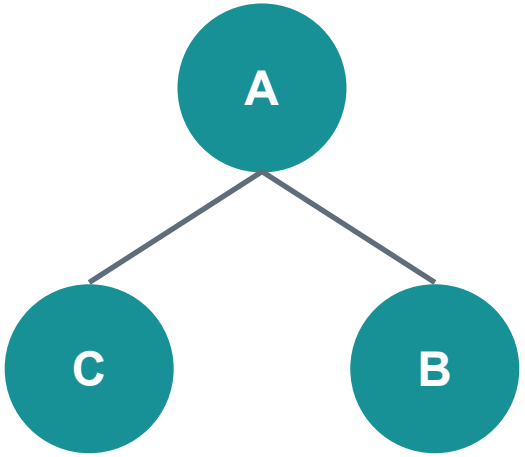
In a binary tree, position matters.

TWO DIFFERENT POSITIONS



B = left child **C = right child**

SWAPPING THEM CHANGES THE TREE



C = left child **B = right child**

Think: Are these two binary trees identical?

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Binary Tree: Basic Properties

These properties will help us reason about size, height and structure.

01 Maximum children

Each node has at most 2 children.

02 Ordered children

Children are distinguished as left and right.

03 Possible child counts

A node can have 0, 1, or 2 children.

04 Recursive structure

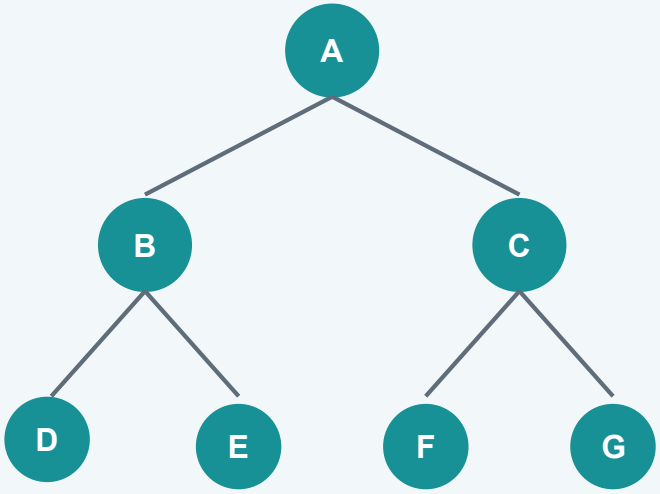
Each child can itself be the root of another binary tree.

Next → Types of Binary Trees: Full, Complete, Perfect and Skewed

Full Binary Tree

Every node has either 0 children or exactly 2 children.

FULL BINARY TREE



Internal nodes → 2 children | Leaves → 0 children

RULE

No node has exactly ONE child.

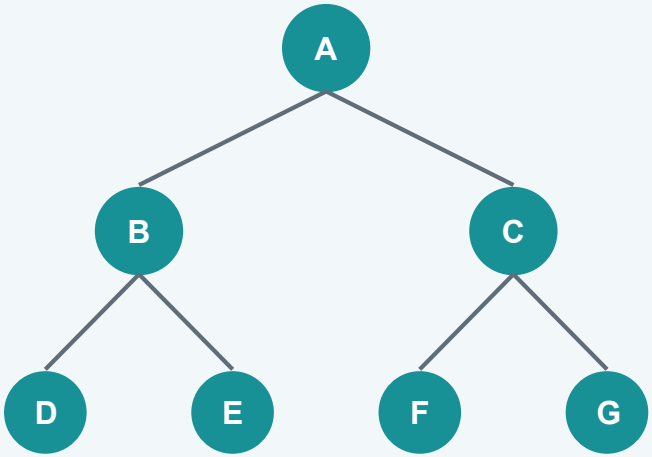
- ✓ 0 children
- ✓ 2 children

Think: Is a single-child node allowed?

Complete Binary Tree

All levels are completely filled except possibly the last; the last level is filled from left to right.

COMPLETE BINARY TREE



Last level: filled from LEFT → RIGHT

KEY TEST

**If the last level has gaps,
the gaps must occur at the RIGHT side.**

Think of filling seats
row by row, left to right.

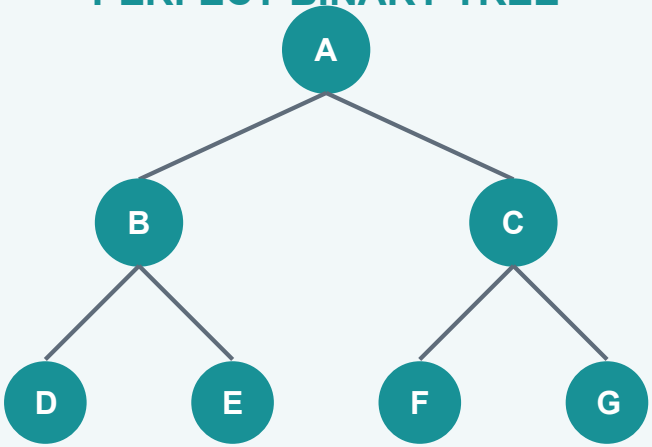
Complete ≠ Perfect

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Perfect Binary Tree

Every internal node has 2 children, and all leaves are at the same level.

PERFECT BINARY TREE



All leaves are at the SAME depth.

QUICK CHECK

✓ Every level is full
✓ Every internal node has 2 children
✓ Leaves are at one level

A perfect tree is always complete.

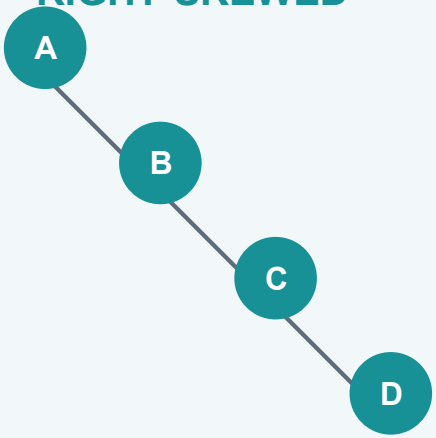
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Skewed Binary Tree

A binary tree in which each node has only one child, producing a chain-like shape.

RIGHT-SKEWED



Each node has only a RIGHT child.

FOUR TYPES — QUICK COMPARISON

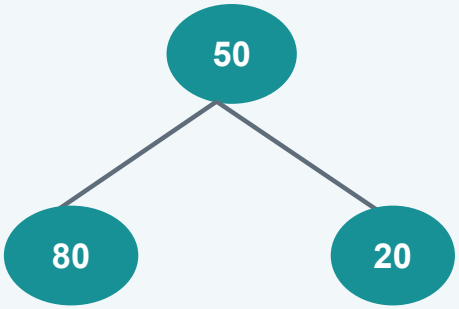
Full	0 or 2 children
Complete	Last level: left → right
Perfect	All levels full
Skewed	One child per node

Next → Identify the type from a given tree

From Binary Tree to Binary Search Tree

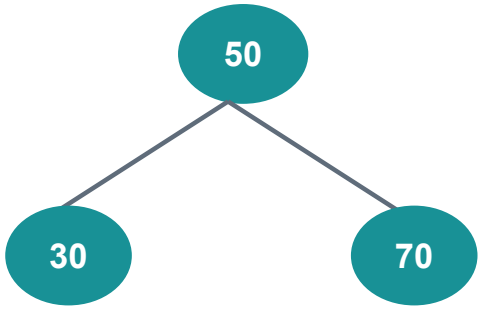
A binary tree gives us structure. A BST adds an ordering rule.

ORDINARY BINARY TREE



The left and right positions exist, but the values need not follow an order.

BINARY SEARCH TREE



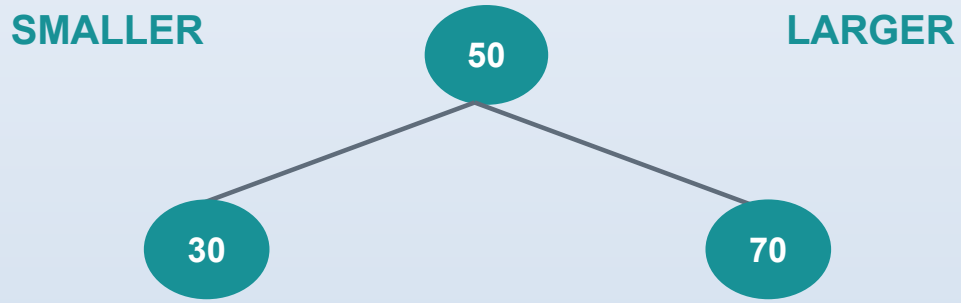
Values follow a rule:
LEFT < ROOT < RIGHT

KEY IDEA → Ordering makes searching faster.

What Is a Binary Search Tree?

BST property is applied recursively to every node.

A Binary Search Tree (BST) is a binary tree in which, for every node, values in the LEFT subtree are smaller and values in the RIGHT subtree are larger.

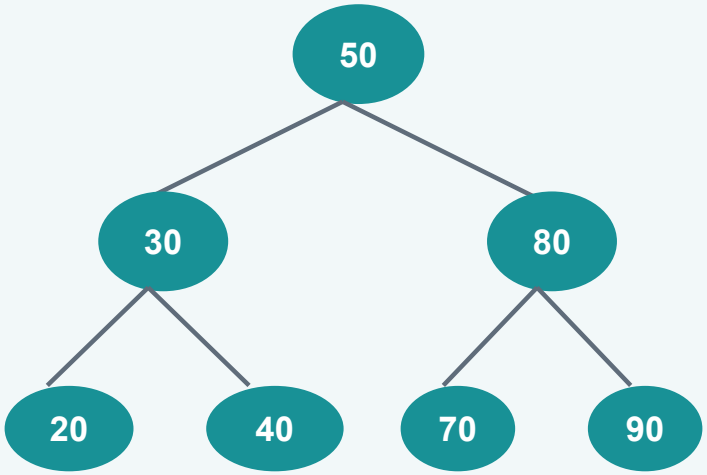


For duplicates, a course must define a consistent policy; here we assume distinct keys.

The BST Rule Applies Everywhere

It is not enough for only the root's children to be ordered.

LOOK AT THE SUBTREES



CHECK ANY NODE

At node 30:
20 < 30 < 40

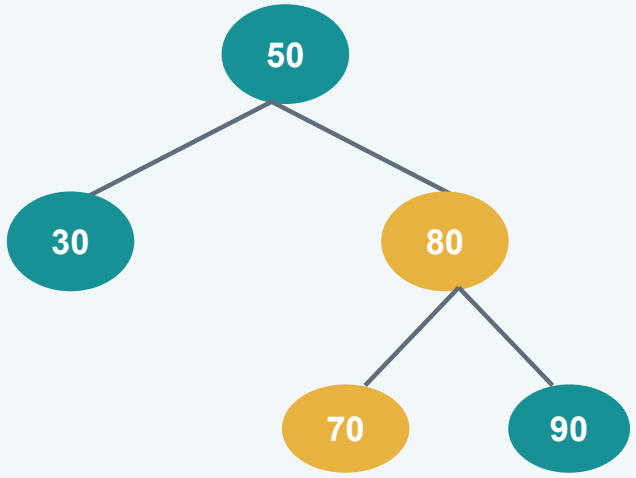
At node 80:
70 < 80 < 90

The same rule holds for EVERY subtree.

Searching in a BST

At each comparison, one subtree can be eliminated.

SEARCH FOR 70



50 → go RIGHT → 80 → go LEFT → 70

THE DECISION

Target < current
→ go LEFT

Target > current
→ go RIGHT

Target = current
→ FOUND

This guided search is the main advantage of the BST ordering.

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Applications of Binary Search Trees

BSTs are useful when data needs to remain ordered while supporting efficient searching and updates.

Fast Searching
Searching for a key by repeatedly choosing the left or right subtree.

Ordered Data
Maintaining a collection where elements can be retrieved in sorted order.

Dynamic Insertions
New values can be inserted while preserving the BST ordering property.

Dynamic Deletions
Values can be removed while keeping the remaining elements organized.

Range Queries
Finding values within a range by using the ordering of the tree.

Symbol Tables
A BST can organize keys used in dictionaries, maps, and lookup structures.

KEY IDEA → BSTs combine hierarchical structure with sorted ordering.

From Arithmetic Expression to Tree

An expression can be represented as a hierarchy of operators and operands.

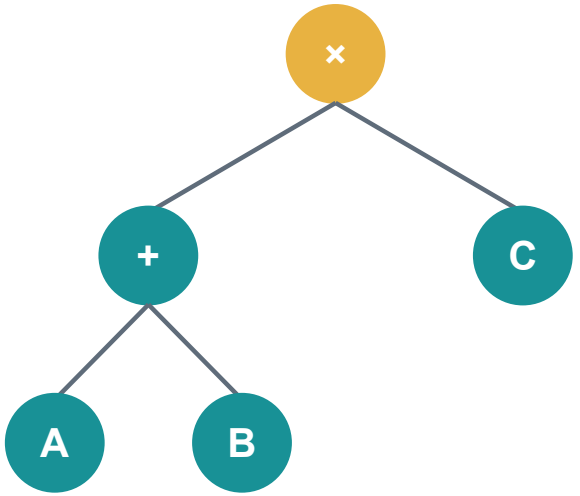
ARITHMETIC EXPRESSION

$$(A + B) \times C$$

The expression has a natural hierarchy:
first A + B, then multiply the result by C.

Can we represent this hierarchy as a tree?

EXPRESSION TREE



Operators → internal nodes
Operands → leaves

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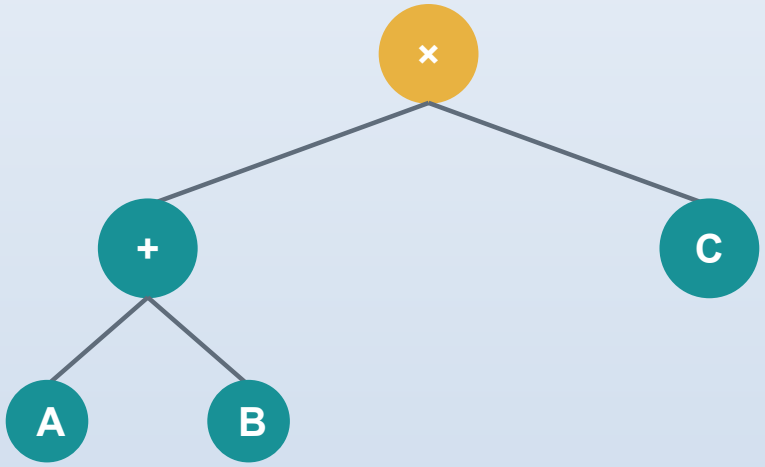
What Is an Expression Tree?

A binary tree used to represent an arithmetic or logical expression.

An expression tree is a binary tree in which OPERATORS are stored in internal nodes and OPERANDS are stored in leaf nodes.

OPERATOR

× and +



OPERANDS

A, B and C

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How Does an Expression Tree Represent Evaluation?

The tree structure shows which operation must be performed first.

EXPRESSION: (A + B) × C

1. **A + B** — Evaluate the left subtree
2. **Result × C** — Use that result with C
3. **Final result** — Expression is evaluated

TREE = HIERARCHY

Root operator

↓

Subexpressions

↓

Operands

The deepest operands are combined first as the expression is evaluated.

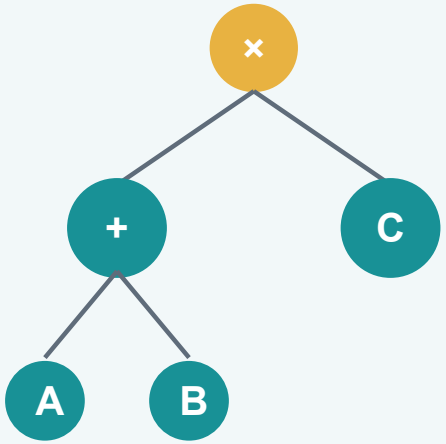
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Expression Tree and Tree Traversals

The same expression can be written in three familiar forms.

ONE TREE → THREE FORMS



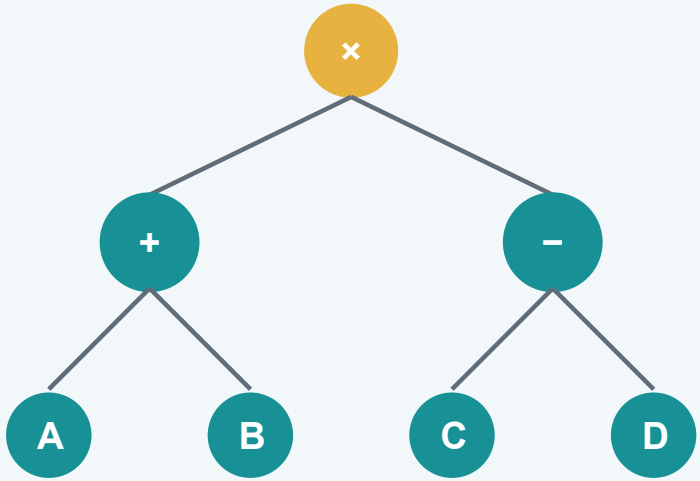
INORDER	**A + B × C**	Infix form
PREORDER	**× + A B C**	Prefix form
POSTORDER	**A B + C ×**	Postfix form

Next → Construct an expression tree from POSTFIX.

Complete Expression Tree Example

Let us build the tree for the expression: (A + B) × (C − D)

EXPRESSION TREE



Operators → ×, +, − **Operands → A, B, C, D**

Root = × Left subtree = (A+B) Right subtree = (C−D)

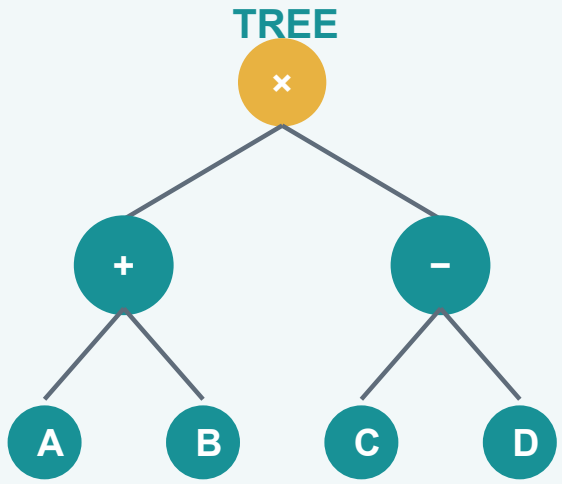
HOW TO READ IT

1. Start at the root.
2. Move through the left and right subtrees.
3. The traversal order determines the expression form.

One tree can produce three expression forms.

Complete Traversal of the Expression Tree

For the same tree, each traversal gives a different expression notation.



INORDER A + B × C − D

Infix

PREORDER × + A B − C D

Prefix

POSTORDER A B + C D − ×

Postfix

Use parentheses in infix to preserve the tree structure: (A+B) × (C−D)

How the Traversals Work

The position of the operator in the traversal determines the notation.

INORDER

Left → Root → Right

A + B × C − D

Operator appears BETWEEN operands

Infix notation

PREORDER

Root → Left → Right

× + A B − C D

Operator appears BEFORE its operands

Prefix notation

POSTORDER

Left → Right → Root

A B + C D − ×

Operator appears AFTER its operands

Postfix notation

Classroom tip → Ask students to trace the tree with a finger while saying the traversal aloud.

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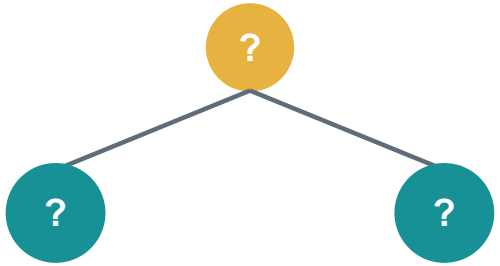
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Your Turn — Build & Traverse

Work out the expression tree and all three traversals.

Expression: (P − Q) + (R × S)

DRAW THE EXPRESSION TREE



Fill in the operators and operands.

FIND THESE

① Inorder →

_____

② Preorder →

_____

③ Postorder →

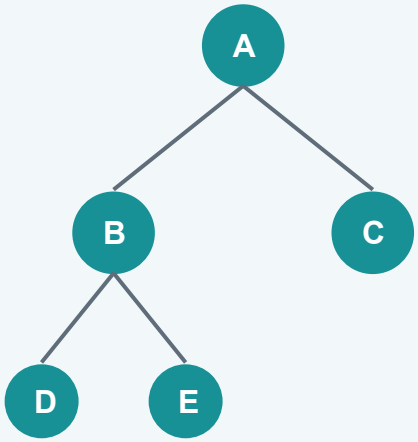
_____

Challenge → Which traversal gives the postfix expression?

From Trees to Forests

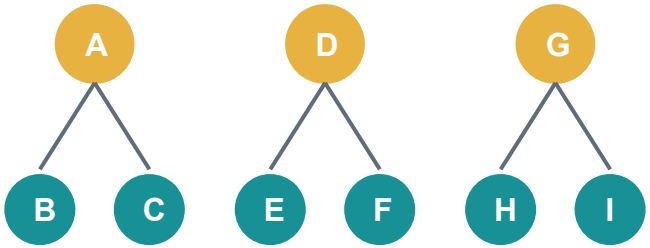
A forest is what we get when we have more than one separate tree.

ONE TREE



Connected structure with one root

A FOREST



Three separate trees = one forest

No single root connects all the trees.

KEY IDEA → A forest is a collection of disjoint trees.

What Is a Forest?

Think of a forest as a set of separate trees.

A forest is a collection of zero or more disjoint trees.

IMPORTANT

- **Each tree has its own root.**
- **Trees do not share nodes.**
- **There is no edge connecting one tree to another.**

IF A FOREST HAS k TREES

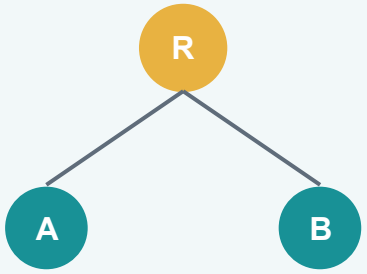
there are k connected components, and k roots.

provided each tree is non-empty.

Tree vs Forest

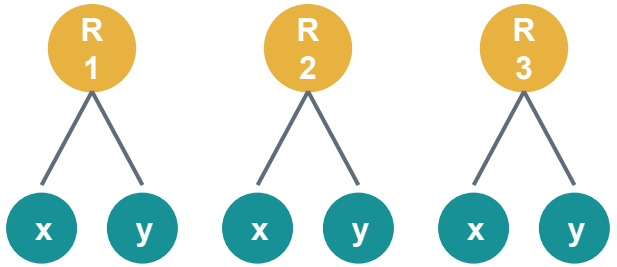
The key difference is whether the components are connected.

TREE



ONE connected component
ONE root

FOREST



THREE connected component
THREE roots

Ask: How many trees can a forest contain?

Properties & Applications of Forests

Forests are useful when a single hierarchy is not enough.

PROPERTIES

✓ A forest may contain multiple trees.

✓ Each tree has its own root.

✓ Trees are disjoint — no shared nodes.

✓ A forest with k trees has k connected components.

APPLICATIONS

Separate organizational hierarchies

Collections of independent file trees

Multiple independent hierarchy structures

Graph algorithms — spanning forests

Next → Forest to Binary Tree conversion

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Important Properties of Trees

These relationships help us analyze the size and structure of a tree.

1 Edges in a tree

A tree with n nodes has exactly n − 1 edges.

2 Unique path

There is exactly one path between any two nodes.

3 Rooted tree

Every node except the root has exactly one parent.

4 Leaf nodes

A leaf is a node with no children.

5 Height & depth

Depth is distance from the root; height is the longest downward path.

6 Subtree

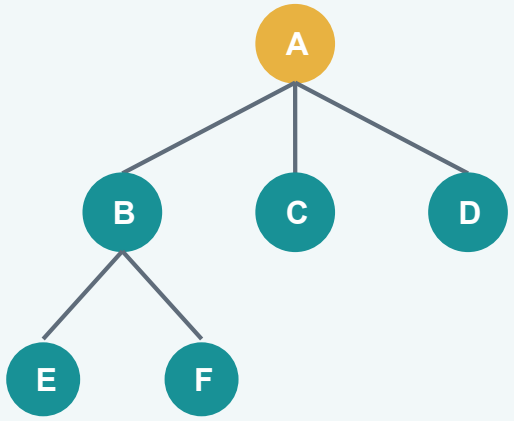
The descendants of a node together with that node form a subtree.

Remember → A tree is connected and contains no cycles.

Converting an N-ary Tree to a Binary Tree

Use the Left-Child / Right-Sibling representation.

N-ARY TREE



A has children B, C, D
B has children E, F

CORE IDEA

LEFT pointer → first child

RIGHT pointer → next sibling

So a node no longer needs many child links.
It uses at most TWO links.

This produces a binary-tree representation.

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N-ary → Binary: Conversion Rules

Remember two simple rules and apply them consistently.

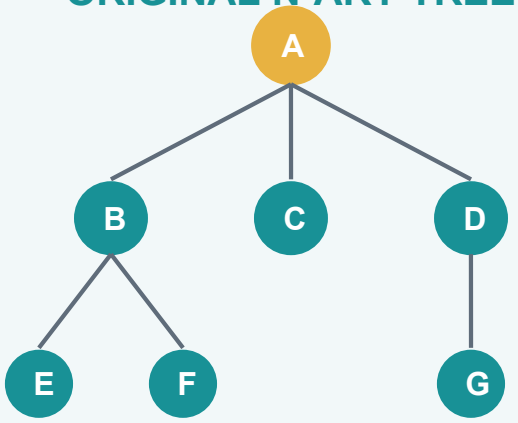
STEP 1	**First child becomes LEFT child**	Connect each node to its first child using the left link.
STEP 2	**Next sibling becomes RIGHT child**	Connect siblings from left to right using right links.
STEP 3	**Remove extra parent-child links**	The remaining child relationships are represented through the sibling chain.

Mnemonic → LEFT = first child | RIGHT = next sibling

Worked Example: N-ary Tree → Binary Tree

Apply Left-Child / Right-Sibling to the same tree.

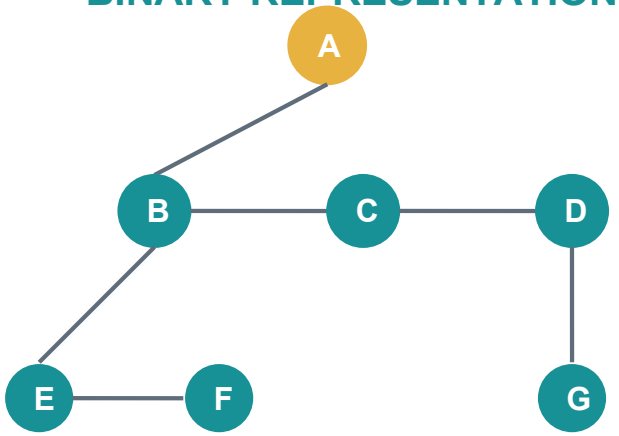
ORIGINAL N-ARY TREE



Children:

A → B, C, D B → E, F D → G

BINARY REPRESENTATION



A.left = B
B.right = C, C.right = D
B.left = E, E.right = F
D.left = G

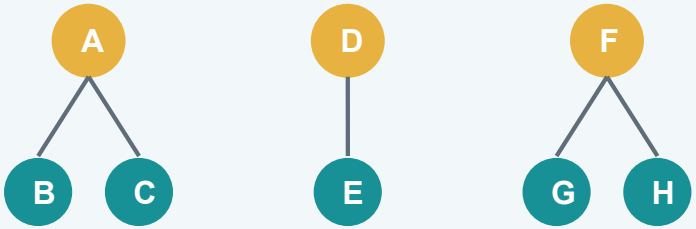
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Converting a Forest to a Binary Tree

The same Left-Child / Right-Sibling idea also represents a forest.

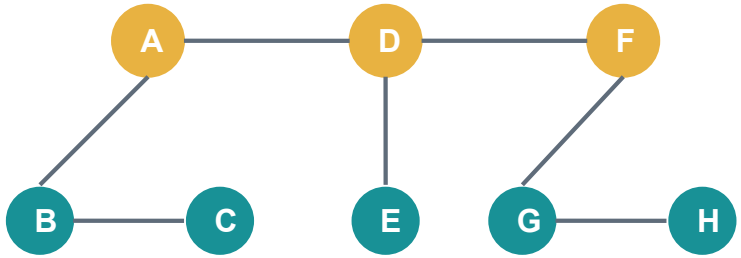
FOREST



Three separate trees
A(B,C) D(E) F(G,H)

Forest roots are treated as siblings.

BINARY REPREPRESENTATION



Forest roots become a RIGHT-SIBLING chain:
A → D → F

Each root's children use LEFT links.

Result: one binary-tree representation.

Thank you !!!!!